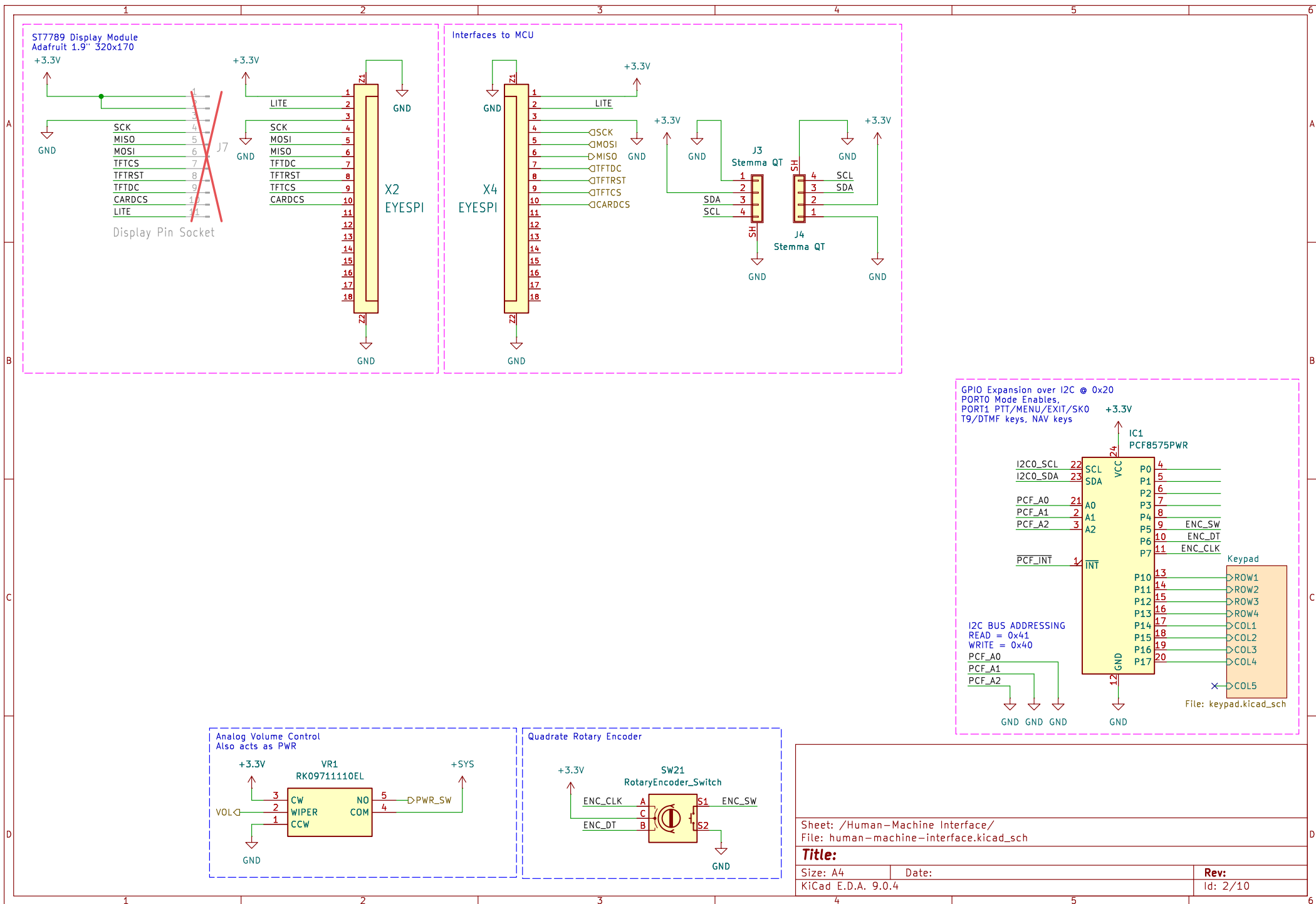
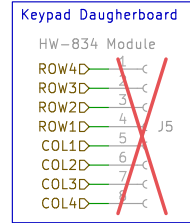
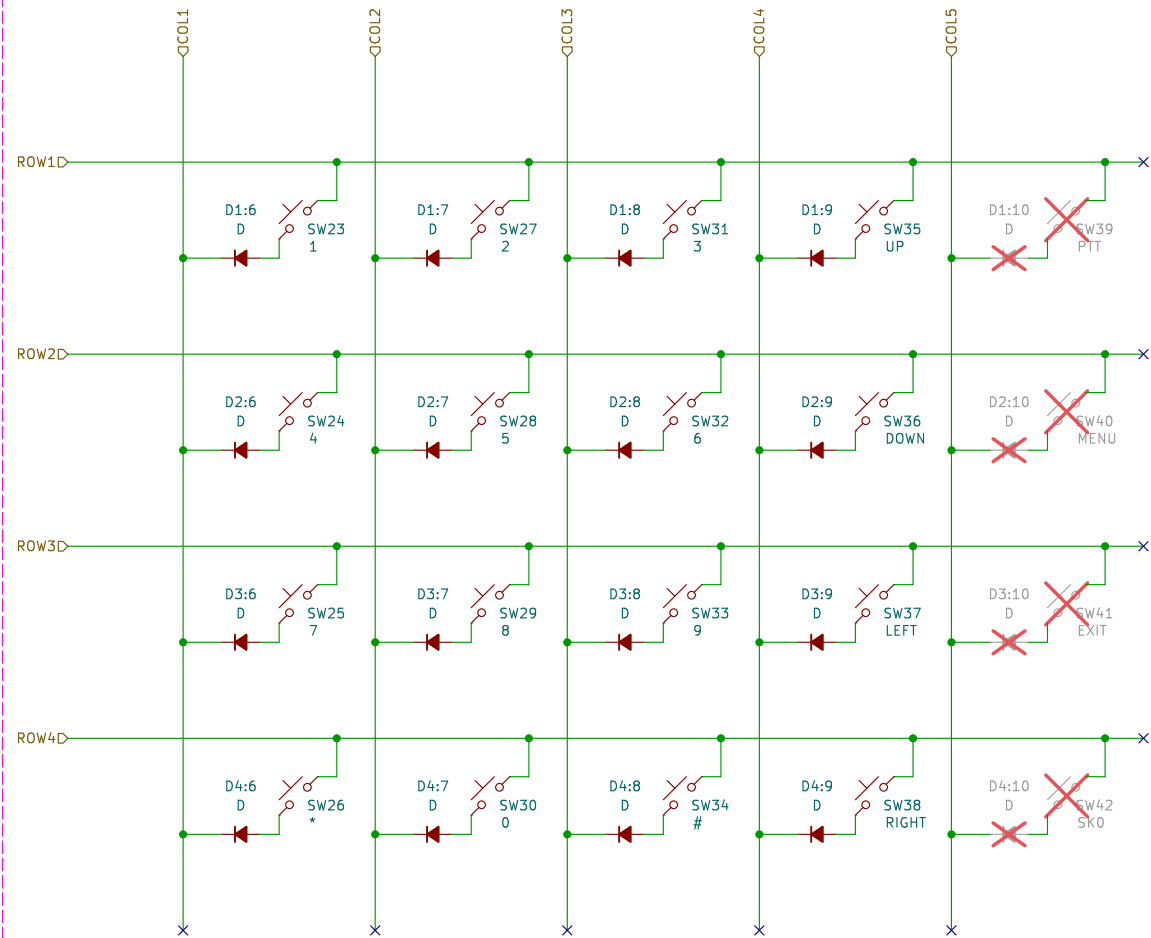




Keypad Matrix
 Application Note: The ROW pins get pulled up by the GPIO expander, with one ROW being pulled down.
 The scanner relies on the COL pins being read in as a logic swing to low.
 Remarked as Domino Logic, a rail-to-rail logic swing occurs.



Sheet: /Human-Machine Interface/Keypad/
 File: keypad.kicad_sch

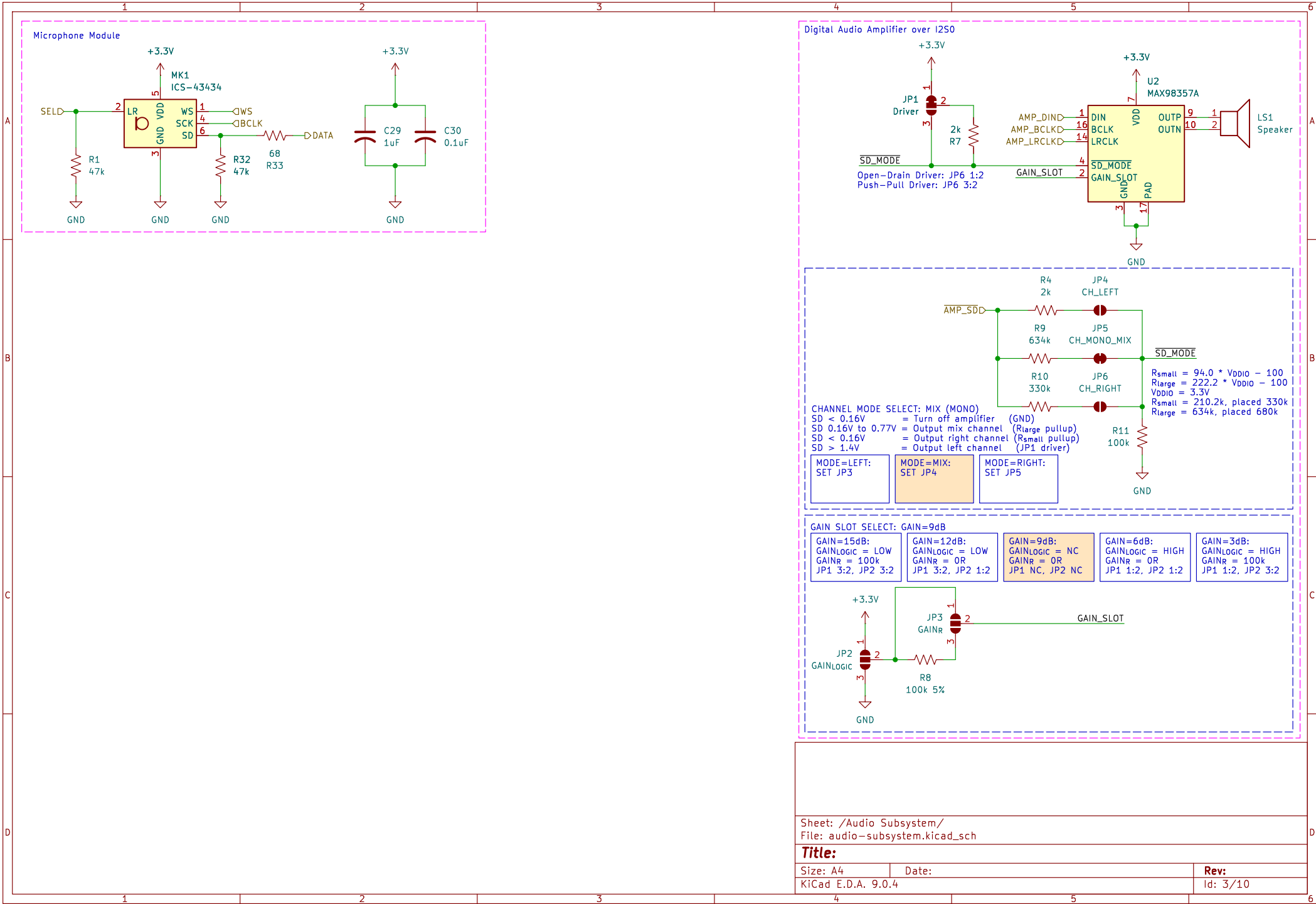
Title:

Size: A4 Date:

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Id: 10/10

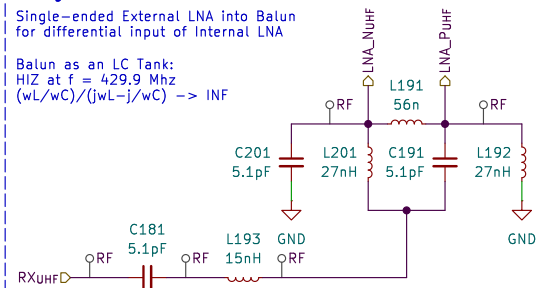


RF Frontend for UHF Reference 420-470 MHz Tune Separated RX and TX Signal Chains

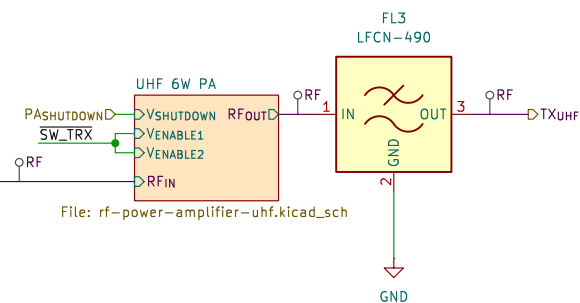
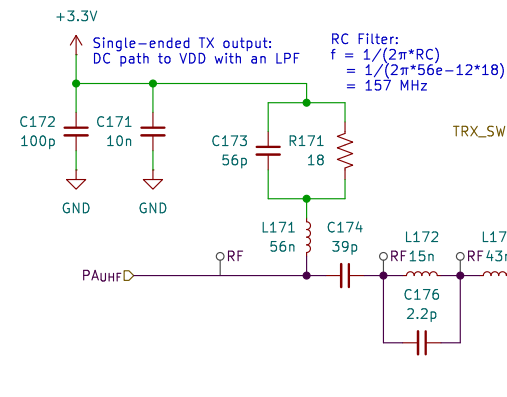
RX Signal Chain

Single-ended External LNA into Balun for differential input of Internal LNA

Balun as an LC Tank:
HIZ at $f = 429.9$ Mhz
(wL/wC)/($jwL-j/wC$) -> INF



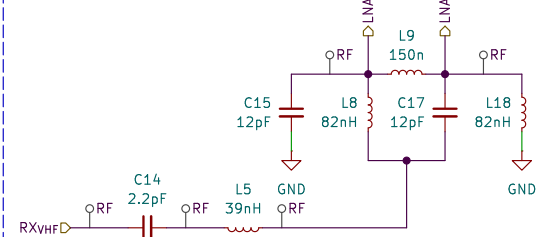
TX Signal Chain



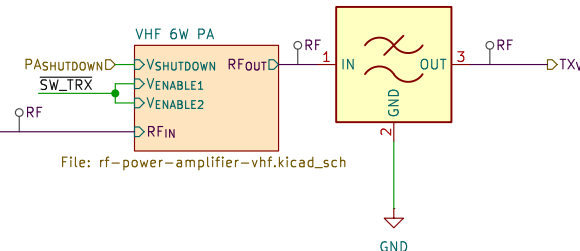
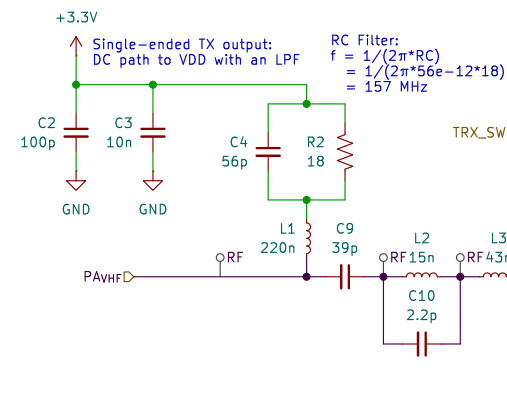
RF Frontend for VHF Reference 169-170 MHz Tune Separated RX and TX Signal Chains

RX Signal Chain

Single-ended External LNA into Balun for differential input of Internal LNA



TX Signal Chain



Sheet: /RF Subsystem/RF Frontend/
File: rf-frontend.kicad_sch

Title:

Size: A4

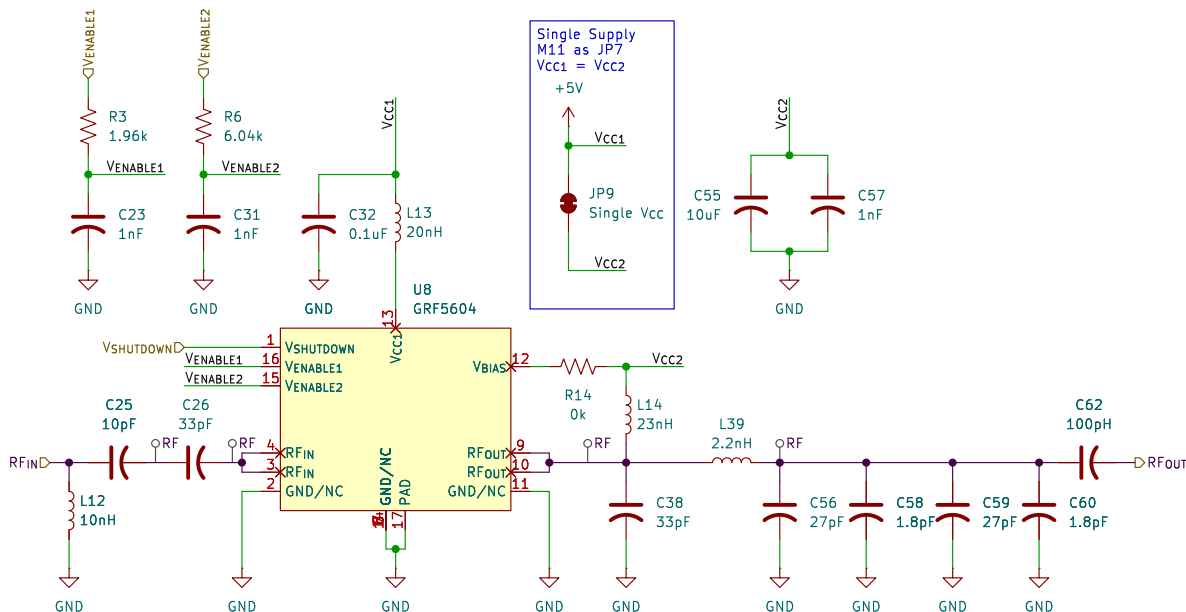
Date:

KiCad E.D.A. 9.0.4

Rev:

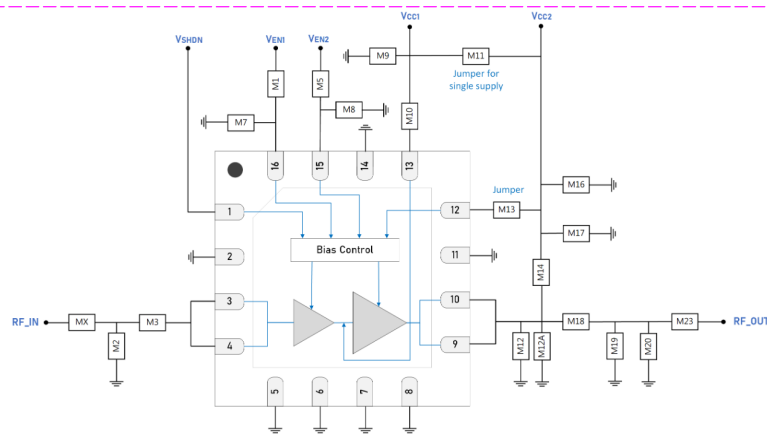
Id: 6/10

RF Amplifier for UHF
Reference 390-470 MHz Tune



Ref	Eval. Ref	450-470 MHz
R	M1	1.96k
L1	M2	10nH
C5	M3	33pF
C3	MX	10pF
R	M5	6.04k
C	M7	1000pF
C	M8	1000pF
C	M9	0.1uF
L	M10	20nH
R	M11	0k
C7	M12	33pF
C	M12A	2.0pF
R	M13	0k
L	M14	23nH
C	M16	**10uF
C	M17	1000pF
L	M18	2.2nH
C8	M19	27pF
C	M20	1.8pF
C	M23	100pF

**10 μ F must be rated for > 5 V at maximum ambient temperature. Manufacturer Part Number in this case = GRM155C80J106ME11D.



GRF5604 Standard Evaluation Board Schematic

Sheet: /RF Subsystem/RF Frontend/UHF 6W PA/
File: rf-power-amplifier-uhf.kicad_sch

Title:

Size: A4

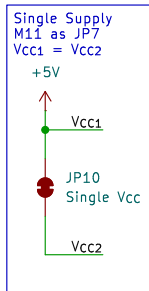
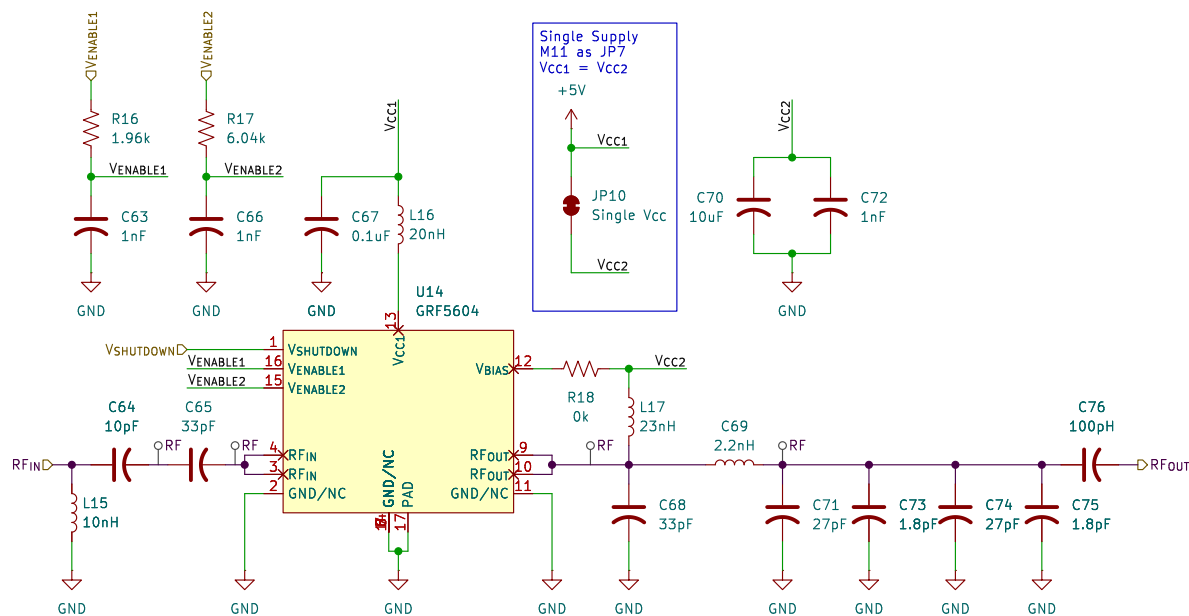
Date:

Rev:

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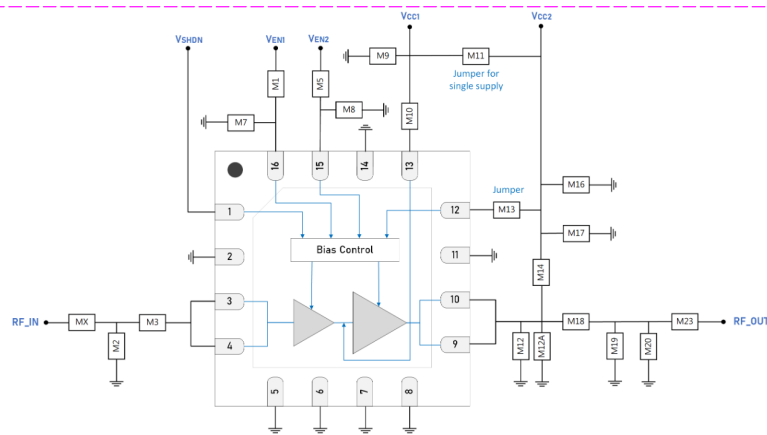
Id: 7/10

RF Amplifier for VHF
Needs unique VHF Tune?



Ref	Eval. Ref	450-470 MHz
R	M1	1.96k
L1	M2	10nH
C5	M3	33pF
C3	MX	10pF
R	M5	6.04k
C	M7	1000pF
C	M8	1000pF
C	M9	0.1uF
L	M10	20nH
R	M11	0k
C7	M12	33pF
C	M12A	2.0pF
R	M13	0k
L	M14	23nH
C	M16	**10uF
C	M17	1000pF
L	M18	2.2nH
C8	M19	27pF
C	M20	1.8pF
C	M23	100pF

**10 uF must be rated for > 5 V at maximum ambient temperature. Manufacturer Part Number in this case = GRM155C80J106ME11D.



GRF5604 Standard Evaluation Board Schematic

Sheet: /RF Subsystem/RF Frontend/VHF 6W PA/
File: rf-power-amplifier-vhf.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.4

Rev:

Id: 8/10

